Calango

Native DFT Engine

All-electron density-functional theory
on numerical atomic orbitals

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Abstract

The Calango Native DFT Engine (`src/dft`) is an all-electron Kohn–Sham solver written directly in C++, using numerical atomic orbitals as its basis. It treats every electron explicitly — no pseudopotential and no frozen core — and runs inside the application rather than by generating a script for an external program.

This document gives the mathematical formulation of the engine: the basis and its generation, the multicentre integration grids, the construction of the overlap and Hamiltonian matrices, the exchange–correlation functionals, Brillouin-zone sampling, the self-consistency loop, and the energy derivatives.

Every section is badged with its implementation status. The ground state is complete and validated; gradient functionals, symmetry reduction of the k-point set, forces and stress are formulated here as *specification* for work not yet done. Section 9 gives the numbers the implemented parts are checked against, and states plainly which quantities have no external reference.

Contents

1 Introduction

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1.1 What the engine is

Two choices define it, and nearly every design decision follows from one of them.

All-electron. Every electron is carried explicitly. This removes the pseudopotential from the list of things a result depends on: a number is wrong only because the functional or the basis is, not because of a transferability assumption made by whoever generated the potential. It is also what makes core-level properties computable at all.

Numerical atomic orbitals (NAO). Basis functions are tabulated on a radial mesh rather than expanded in analytic primitives, so a basis function can be the *exact numerical solution* of the atomic problem. The free atom is therefore essentially exact at the smallest possible basis size, and each further level has a physical meaning. Confinement makes each function *exactly* zero beyond a finite radius, which is what turns the overlap and Hamiltonian matrices sparse.

The price is paid in the integrals: nothing is analytic. Every matrix element is a numerical quadrature over a real-space grid, so *the grid is the accuracy*, and most of the engineering lives in building grids that are dense where the wavefunction has structure and sparse where it does not.

1.2 Units and conventions

Hartree atomic units throughout ($\hbar = m_e = e = 4\pi\varepsilon_0 = 1$); lengths in bohr, energies in hartree. The application boundary converts to Ångström and electronvolt, once. Spherical harmonics are the *real* $Y_{\ell m}$, normalised so $\int |Y_{\ell m}|^2 \mathrm{d}\Omega = 1$, without the Condon–Shortley phase.

1.3 Implementation status

Capability	Section	Status
Radial atomic solver, basis generation	2	implemented
Multicentre integration grids	3	implemented
Overlap, kinetic and potential matrices	4	implemented
Electrostatics (neutral-atom + multipole)	4.2	implemented
LDA exchange–correlation	5.1	implemented
GGA (PBE)	5.2	implemented
Density gradients $\nabla\rho$ on the grid	5.2	implemented
Monkhorst–Pack mesh, Bloch sums, $H(\mathbf{k})C = S(\mathbf{k})CE$	6	implemented
Irreducible-zone symmetry reduction	6.3	implemented
— exact folding (time reversal)	6.3	implemented
— density symmetrisation	6.3	**not implemented**
SCF loop, linear and Pulay mixing	7	implemented
Forces (finite difference) and relaxation	8.1	implemented
— analytic Pulay, full	8.1	**not implemented**
Stress tensor	8.2	implemented
Spin polarisation	—	**not implemented**

The engine reports the same list at run time from `CalangoDFTEngine::unimplementedSteps()`, so a user interface cannot drift from it. Nothing in the engine returns a number it did not compute: a step that cannot run reports `NotImplemented` and leaves its output untouched.

2 Basis sets

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2.1 Form of a numerical atomic orbital

A basis function centred on nucleus A is a tabulated radial function times a real spherical harmonic,

$$\phi_{Ai}(\mathbf{r}) = \frac{u_{n\ell}(r_A)}{r_A} Y_{\ell m}(\hat{\mathbf{r}}_A), \qquad \mathbf{r}_A = \mathbf{r} - \mathbf{R}_A. \tag{1}$$

The radial part is stored as $u = rR$ rather than R. That is not a bookkeeping detail: u is what the radial Schrödinger equation is written in,

$$u'' = \left[\frac{\ell(\ell+1)}{r^2} + 2\big(v(r) - \varepsilon\big)\right] u, \tag{2}$$

with no first-derivative term; u vanishes at the origin for every ℓ, so the nuclear cusp is not a numerical problem; and $\int_0^\infty |u|^2 \mathrm{d}r = 1$ is the normalisation, with no r^2 weight to lose precision on.

2.2 The logarithmic radial mesh

Every all-electron quantity is dominated by a region a fraction of a bohr across. The $1s$ density of a third-row element lives inside $0.1\,a_0$ while the valence tail runs to ten. The mesh is therefore

$$r(i) = a\left(e^{bi} - 1\right), \qquad i = 0, \ldots, N-1, \tag{3}$$

which starts *exactly* at $r = 0$ — so the cusp sits on a grid point rather than between two — and turns every radial integral into a uniformly spaced one in i:

$$\int f(r)\mathrm{d}r = \int f\big(r(i)\big)\frac{\mathrm{d}r}{\mathrm{d}i}\mathrm{d}i, \qquad \frac{\mathrm{d}r}{\mathrm{d}i} = b\,(r+a). \tag{4}$$

Simpson's rule applies directly in i. Two consequences are used throughout: interpolation is done in the *index*, where the mesh is uniform and a cubic is well conditioned everywhere, and the index for a given radius is obtained by inverting (3) in closed form, $i = \ln(1 + r/a)/b$, rather than by searching.

Running integrals use *cumulative* Simpson, not partial sums of the composite weights: the $(1, 4, 2, \ldots, 4, 1)/3$ pattern is a valid quadrature only for the whole interval, and truncating it leaves a 4 or a 2 where an endpoint 1 belongs — an $\mathcal{O}(h)$ error at every point instead of $\mathcal{O}(h^4)$.

2.3 The free atom

Basis generation starts by solving the spherical, non-spin-polarised free atom self-consistently. Equation (2) is integrated as a first-order pair $(u, \mathrm{d}u/\mathrm{d}r)$ in the mesh index by classical Runge–Kutta.

The eigenvalue is found by **bisection on the node count**. The number of nodes of the outward solution is a non-decreasing step function of ε that jumps by one at each eigenvalue, so bracketing the jump brackets the eigenvalue — with no derivative, no matching condition, and no way to converge to a neighbouring state however close it lies. The wavefunction is then rebuilt by joining an outward solution to an inward one at the classical turning point, because beyond that point the outward solution is the *growing* one and what it actually carries is rounding error amplified exponentially.

2.4 Confinement

Strict locality requires each function to vanish identically beyond a cutoff. This is imposed as a *finite potential barrier added to the potential the orbital is solved in*, of the form of Junquera *et al.* [3]:

$$v_{\mathrm{conf}}(r) = \begin{cases} 0, & r \le r_i, \\ V_0 \, \dfrac{\exp[-(r_c - r_i)/(r - r_i)]}{r_c - r}, & r_i < r < r_c, \\ \infty, & r \ge r_c, \end{cases} \tag{5}$$

with onset $r_i = r_c - w$. It is zero inside the onset, and every derivative is continuous where it switches on. In practice the barrier is capped at a finite height, because the radial integrator takes a fixed step in the mesh index and an uncapped barrier drives $\sqrt{2v}\,\Delta r$ outside the stability region of the integrator.

Why a barrier and not a wall. This is worth stating precisely, because the obvious alternative — solving the atom in a box, $u(r_c) = 0$ — is wrong in a way that is almost invisible. A hard wall gives $u(r_c) = 0$ with $u'(r_c) \neq 0$. The orbital is then continuous at the cutoff sphere while its gradient jumps, so $\nabla^2\phi$ contains a *surface delta function* on that sphere. The engine stores the kinetic operator per function as $(\varepsilon - v)u$ (Section 4.1), which is built from the smooth part of u'' alone and cannot represent that delta.

What that costs is hidden everywhere except one place. Diagonal elements are untouched, because the delta is weighted by $u(r_c) = 0$ — so self-consistency, electron counts and kinetic-energy identities all still pass. Every *inter-atomic* element is short by a surface term, because a neighbour's orbital is not zero on this atom's cutoff sphere. The observable signature is exactly that: isolated atoms exact, two atoms far apart exact, and an error that grows with coordination and with the number of cutoff spheres in the basis. Table 9 in Section 9 gives the measured size.

A finite barrier removes the delta instead of approximating it: v stays finite everywhere, so $-\frac{1}{2}\nabla^2\phi = (\varepsilon - v)\phi$ holds at *every* radius, and u reaches the cutoff with both it and its derivative already damped away.

2.5 Basis tiers

The basis is built in nested levels, each strictly containing the last.

Tier 1 (minimal). The confined atom's own occupied orbitals — all of them, core included, because this is an all-electron method and no pseudopotential stands in for the core.

Tier 2 (double zeta plus polarisation). A second radial function for each *valence* angular momentum, plus one shell at the first ℓ the free atom does not occupy (d for silicon). The polarisation function is a genuine solution of (2) in the confined atom's potential: unbound in the free atom, bound in the barrier.

The second zeta is the *split-valence* construction, which needs no bonded reference calculation and no fitting. A radius r_s is chosen so that a fixed fraction of the orbital's norm lies beyond it; inside r_s the orbital is replaced by a smooth polynomial, and the new basis function is the difference:

$$u^{(2\zeta)}(r) \;=\; \begin{cases} u(r) - r^{\ell+1}(a - br^2 + cr^4), & r < r_s, \\ 0, & r \geq r_s. \end{cases} \tag{6}$$

The three coefficients match the orbital in **value, slope and curvature** at r_s. Two parameters — matching value and slope only — is the form usually quoted, and it is not sufficient here for the same reason a hard wall is not: $-\frac{1}{2}\nabla^2\phi$ then jumps at r_s, and a multicentre grid with a few tens of radial shells cannot integrate a discontinuity. Matching the curvature makes u, u' and u'' all vanish at r_s, so the kinetic operator goes smoothly to zero.

The kinetic energy of the added function stays analytic. For $F = r^\ell(a - br^2 + cr^4)$ the $r^{\ell-2}$ terms of the Laplacian cancel against the centrifugal term exactly, leaving

$$-\tfrac{1}{2}\nabla^2[F(r)Y_{\ell m}] \;=\; \Big[b(2\ell+3) - 2c(2\ell+5)r^2\Big] r^\ell\, Y_{\ell m}. \tag{7}$$

Tier 3. A third zeta for the valence channels and a second zeta on the polarisation channel.

3 Integration grids

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There is no analytic integral anywhere in the engine, so the grid *is* the accuracy of every number it produces. Its construction answers one question: how do you integrate a function with a cusp at every nucleus and smooth behaviour in between, without a mesh fine enough for the cusps everywhere?

3.1 Multicentre partition

Becke's answer [1], which the engine implements: give every atom its own spherical grid, dense at its own nucleus, and multiply each by a smooth weight $w_A(\mathbf{r})$ with

$$\sum_A w_A(\mathbf{r}) = 1 \;\text{ everywhere}, \qquad w_A \approx 1 \text{ near } \mathbf{R}_A, \quad w_A \approx 0 \text{ near any other nucleus.} \tag{8}$$

Then $\int f = \sum_A \int w_A f$, and each term integrates a function that is smooth except at *its own* centre — exactly what a spherical grid is good at. The weight is built from the elliptical coordinate between each pair of centres,

$$\mu_{AB} = \frac{r_A - r_B}{R_{AB}}, \tag{9}$$

smoothed by $p(x) = \frac{1}{2}(3x - x^3)$ iterated three times,

$$s(\mu) = \tfrac{1}{2}\Big[1 - p\big(p(p(\mu))\big)\Big], \qquad w_A = \frac{\prod_{B\neq A} s(\nu_{AB})}{\sum_C \prod_{B\neq C} s(\nu_{CB})}, \tag{10}$$

where ν is μ shifted by the atoms' Bragg–Slater radii so the boundary between a large atom and a small one sits where the density says it should.

For a periodic system the partition must see the atoms' *periodic images*, or the weights stop summing to one at the cell boundary and the integrated electron count comes out short by whatever charge lives there.

3.2 Radial and angular rules

Radial shells use Gauss–Chebyshev quadrature of the second kind under the map $r = R(1 + x)/(1 - x)$, which sends a finite interval to the half line, so a fixed number of nodes covers the cusp and the tail at once.

Angular points use Lebedev rules [2] — the octahedrally symmetric optimum, exact for all spherical harmonics up to a given degree with roughly half the points a product rule needs — with a generated Gauss–Legendre × uniform-φ product rule above the tabulated orders. Both are verified rather than trusted: the test suite integrates $\int Y_{\ell m} Y_{\ell' m'} \mathrm{d}\Omega$ on each rule and requires $\delta_{\ell\ell'}\delta_{mm'}$ to machine precision. A mistyped Lebedev weight is otherwise completely silent.

Angular resolution is set by the partition, not the density. The Becke weight has a sharp boundary between atoms, and it is harder to integrate than anything else on the grid. Integrating unity over a silicon primitive cell — which must return the cell volume — is in error by 10% at degree 11 and reaches 5×10^{-5} only at degree 29.

4 Hamiltonian and matrix assembly

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With Bloch sums $\chi_i^{\mathbf{k}}$ defined in Section 6, every matrix element is a quadrature over the multicentre grid:

$$S_{ij}(\mathbf{k}) = \sum_g w_g\, \chi_i^{\mathbf{k}*}(\mathbf{r}_g)\, \chi_j^{\mathbf{k}}(\mathbf{r}_g), \tag{11}$$

$$H_{ij}(\mathbf{k}) = \sum_g w_g\, \chi_i^{\mathbf{k}*}(\mathbf{r}_g) \Big[\hat{T} + v_{\text{eff}}(\mathbf{r}_g)\Big] \chi_j^{\mathbf{k}}(\mathbf{r}_g). \tag{12}$$

4.1 Kinetic energy

The kinetic term is **never differenced**. Each basis function solves the radial equation of the confined atom it came from, so

$$-\tfrac{1}{2}\nabla^2\phi_i \;=\; \big(\varepsilon_i - v_{\text{at}}(r)\big)\phi_i \tag{13}$$

holds identically, and $\hat{T}$ acting on a basis function is a *multiplication* by a tabulated function. Written the obvious way — a numerical Laplacian, or the symmetric $\boldsymbol{\nabla}\phi_i \cdot \boldsymbol{\nabla}\phi_j$ form — the largest single term in the total energy would carry the noise of one or two numerical derivatives of an interpolated table.

Equation (13) requires v_{at} to be the *full* potential the orbital solves, confining barrier included (Section 2.4). Each function therefore carries its own tabulated array

$$t_i(r) \;\equiv\; r \cdot (-\tfrac{1}{2}\nabla^2\phi_i)/Y_{\ell m} \;=\; (\varepsilon_i - v_{\text{at}}(r))u_i(r), \tag{14}$$

which is smooth through the origin (it behaves as Zr^{ℓ}, the $1/r$ of the nucleus being cancelled by the $r^{\ell+1}$ of u). Carrying the array rather than the eigenvalue is what allows basis functions that are *not* eigenstates of anything — a split-valence zeta is an orbital minus a polynomial, and its kinetic array comes from (7). The identity also inverts,

$$u'' = \frac{\ell(\ell+1)}{r^2}u - 2t, \tag{15}$$

so a function's curvature is available without any numerical second derivative.

The matrix is symmetrised, $\frac{1}{2}[\langle\phi_i|\hat{T}\phi_j\rangle + \langle\hat{T}\phi_i|\phi_j\rangle]$, because the identity holds with each function's own ε and v_{at}.

4.2 Electrostatics

The total charge of the system — electrons plus nuclei — is written as a sum of *neutral* free atoms plus a difference density:

$$n(\mathbf{r}) \;=\; \sum_{A,\mathbf{T}} \Big[\rho_A^{\text{free}}(\mathbf{r} - \mathbf{R}_A - \mathbf{T}) - Z_A\,\delta(\mathbf{r} - \mathbf{R}_A - \mathbf{T})\Big] \;+\; \delta\rho(\mathbf{r}). \tag{16}$$

Each bracket is neutral, so its potential decays *exponentially* instead of as $1/r$ and the lattice sum converges absolutely — no Ewald summation, no conditional convergence, no compensating background. The neutral-atom potential is

$$v_A^{\text{NA}}(r) \;=\; -\frac{Z_A}{r} + v_H[\rho_A^{\text{free}}](r). \tag{17}$$

In the engine the nuclear $1/r$ is subtracted analytically at each point and only the screened remainder is interpolated, because interpolating v^{NA} itself means fitting a cubic across a pole at the one place the density is largest.

What is left, $\delta\rho$, is the small bonding rearrangement, and it integrates to zero over the cell. Its potential is solved centre by centre. The grid points of atom A already carry the Becke weight, so they *are* $w_A\delta\rho$ sampled on a spherical grid; expanding in real spherical harmonics,

$$c_{\ell m}^A(r) = \int \delta\rho_A(r,\Omega)\,Y_{\ell m}(\Omega)\mathrm{d}\Omega, \tag{18}$$

the radial Poisson equation has the closed solution

$$V_{\ell m}^A(r) \;=\; \frac{4\pi}{2\ell+1}\left[\frac{1}{r^{\ell+1}}\int_0^r c_{\ell m}^A(r')\,r'^{\,\ell+2}\mathrm{d}r' + r^{\ell}\int_r^\infty c_{\ell m}^A(r')\,r'^{\,1-\ell}\mathrm{d}r'\right], \tag{19}$$

continued outside the sphere by its exact multipole tail. The total electrostatic potential and energy are then

$$v_{\text{es}}(\mathbf{r}) = \sum_{A,\mathbf{T}} v_A^{\text{NA}}(|\mathbf{r} - \mathbf{R}_A - \mathbf{T}|) + v_H[\delta\rho](\mathbf{r}), \tag{20}$$

$$E_{\text{es}} = E_{\text{ref}} + \int \delta\rho \sum_{A,\mathbf{T}} v_A^{\text{NA}} + \tfrac{1}{2}\int \delta\rho\; v_H[\delta\rho], \tag{21}$$

where E_{ref} — each free atom's own electrostatic self-energy plus every neutral-atom pair interaction — depends only on the geometry and is computed once. Note that $v_{\text{es}} = \delta E_{\text{es}}/\delta\rho$ exactly, which is what keeps the energy and the Hamiltonian consistent.

Known limitation. The monopole tail of $\delta\rho$ is truncated. That is exact when each atom's difference density is neutral, which symmetry guarantees for a crystal whose atoms are all equivalent, and an approximation otherwise. The engine computes the per-atom monopole regardless and reports it when it is not negligible, rather than absorbing it silently.

5 Exchange–correlation functionals

5.1 Local density approximation

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Exchange is Dirac–Slater, which is not a parameterisation of anything — it is the exact exchange energy density of the uniform electron gas:

$$\varepsilon_x(\rho) = -\frac{3}{4}\left(\frac{3\rho}{\pi}\right)^{1/3}, \qquad v_x = \frac{\partial(\rho\varepsilon_x)}{\partial\rho} = \frac{4}{3}\varepsilon_x. \tag{22}$$

That factor of 4/3 is exact — $\rho\varepsilon_x$ scales as $\rho^{4/3}$ — and is the cheapest available check that an exchange implementation is right.

Three correlation parameterisations of the same Ceperley–Alder quantum Monte Carlo electron gas are available, written in $r_s = (3/4\pi\rho)^{1/3}$: Perdew–Zunger [4], Vosko–Wilk–Nusair (formula V) [5], and Perdew–Wang 1992 [6], which is the default. PW92 is the default for a practical reason: it is what most plane-wave codes mean by "LDA" without qualification, so a number computed with it can be compared against theirs to every digit the grids support, rather than to the few meV per electron by which the fits differ. Its form is a single analytic expression over the whole density range,

$$\varepsilon_c(r_s) = -2A(1+\alpha_1 r_s)\,\ln\left[1 + \frac{1}{2A(\beta_1 r_s^{1/2} + \beta_2 r_s + \beta_3 r_s^{3/2} + \beta_4 r_s^2)}\right], \tag{23}$$

with the potential following from $v_c = \varepsilon_c - \frac{r_s}{3}\,\mathrm{d}\varepsilon_c/\mathrm{d}r_s$, taken analytically and verified against a finite difference of $\rho\varepsilon_c$ in the test suite.

Energy and potential are computed and returned *separately*. Computing the energy as $\int \rho\, v_{xc}$ — a mistake that survives an SCF, because the density still converges — overestimates the exchange energy by exactly one third.

5.2 Generalised gradient approximation (PBE)

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A GGA is written in the density and the square of its gradient, $\sigma = |\nabla\rho|^2$, as an energy per unit *volume* $f(\rho,\sigma)$ — there is no natural per-electron split once a gradient term is present. The engine returns f, $\partial f/\partial\rho$ and $\partial f/\partial\sigma$ from one interface for both families; an LDA simply has $\partial f/\partial\sigma \equiv 0$, so the assembler needs one code path.

PBE exchange multiplies the uniform-gas energy density by

$$F_x(s) = 1 + \kappa - \frac{\kappa}{1+\mu s^2/\kappa}, \qquad s = \frac{|\nabla\rho|}{2k_F\rho}, \quad k_F = (3\pi^2\rho)^{1/3}, \tag{24}$$

with $\kappa = 0.804$ fixed by the Lieb–Oxford bound and $\mu = \beta\pi^2/3$ chosen to recover the gradient expansion of the linear *response*. Everything is implemented in $t = s^2$ rather than s, so no square root of σ appears: the derivatives stay rational and $\sigma = 0$ is an ordinary point rather than one where $\partial s/\partial\sigma$ diverges.

Correlation adds to the PW92 local part the gradient term

$$H = \gamma \ln\left[1 + \frac{\beta}{\gamma}\,\frac{T(1+AT)}{1+AT+A^2T^2}\right], \qquad A = \frac{\beta/\gamma}{e^{-\varepsilon_c^{\mathrm{unif}}/\gamma} - 1}, \tag{25}$$

with $T = t^2$, $t = |\boldsymbol{\nabla}\rho|/(2k_s\rho)$ and $k_s = \sqrt{4k_F/\pi}$. Note that A depends on the density *through* ε_c, so $\partial f/\partial\rho$ carries a term through A that is easy to omit; it is included, and the derivative is confirmed against a finite difference of the energy rather than trusted. As $\varepsilon_c \to 0^-$ in the low-density tail A diverges while $H \to 0$; the limit is taken explicitly instead of passing through an intermediate infinity.

5.2.1 Density gradients

$\boldsymbol{\nabla}\rho$ is analytic, not differenced. With the density matrix D_{ij},

$$\rho = \sum_{ij} D_{ij}\phi_i\phi_j, \qquad \boldsymbol{\nabla}\rho = 2\sum_{ij} D_{ij}\,\phi_i\,\boldsymbol{\nabla}\phi_j, \tag{26}$$

and the gradient of a single orbital follows from writing it as $\phi = g(r)\,S_{\ell m}$ with $S_{\ell m} = r^\ell Y_{\ell m}$ the real *solid* harmonic — a homogeneous polynomial:

$$\boldsymbol{\nabla}\phi = \left(R' - \frac{\ell R}{r}\right)\hat{\mathbf{r}}\,Y_{\ell m} + \frac{R}{r^\ell}\,\boldsymbol{\nabla}S_{\ell m}. \tag{27}$$

Both terms are finite everywhere. The obvious alternative — differentiating $Y_{\ell m}$ in spherical angles — carries a $1/\sin\theta$ and is singular at the poles.

The radial derivative $\mathrm{d}u/\mathrm{d}r$ is tabulated once per basis function with a *five*-point stencil in the mesh index. The three-point form is not enough: a split-valence function is already the difference of two similar quantities and loses digits to cancellation before any stencil error is added, which left $\boldsymbol{\nabla}\phi$ about a percent off for exactly those functions while the ordinary orbitals were correct to a few parts in ten thousand.

5.2.2 The matrix element

The GGA potential contains $\boldsymbol{\nabla}\cdot(\partial f/\partial\boldsymbol{\nabla}\rho)$, and taking a divergence numerically on an unstructured multicentre grid is both awkward and noisy. The matrix element is therefore integrated by parts:

$$\langle\phi_i|v_{xc}|\phi_j\rangle = \int \frac{\partial f}{\partial\rho}\,\phi_i^*\phi_j\mathrm{d}\mathbf{r} + \int \mathbf{V}\cdot(\phi_i^*\boldsymbol{\nabla}\phi_j + \phi_j\boldsymbol{\nabla}\phi_i^*)\,\mathrm{d}\mathbf{r}, \qquad \mathbf{V} = 2\frac{\partial f}{\partial\sigma}\boldsymbol{\nabla}\rho. \tag{28}$$

This needs only $\boldsymbol{\nabla}\phi$ — no second derivatives, no divergence of a sampled vector field — and is manifestly Hermitian.

One consequence propagates into the energy. The kinetic energy can no longer be obtained as the band energy minus a potential trace, because that form assumes v_{eff} is a local multiplier and the second term of (28) is not. T_s is instead taken directly as the expectation value of $\hat{T}$ over the occupied orbitals, which is exact for both families and reproduces the local result unchanged.

5.2.3 Validation

Checked in three independent ways.

Against arithmetic: $\partial f/\partial\rho$ and $\partial f/\partial\sigma$ agree with finite differences of f to 2×10^{-10} across five decades of density and gradient, and at $\sigma=0$ PBE reduces to LDA-PW92 to 10^{-16} — exactly, as its construction requires. Equation (27) agrees with a finite difference of ϕ to 4×10^{-4} for every function of a double-zeta-plus-polarisation basis.

Against another implementation: f and $\partial f/\partial\rho$ agree with an independent PBE kernel to 10^{-6} relative.

Against physics: the whole pipeline shifts the silicon atom's total energy by −27.37 eV on going from LDA to PBE, against −27.46 eV from an independent all-electron atomic solver — the 0.09 eV residue being the confinement, not the functional. The cohesive energy of bulk silicon moves from 5.68 to 5.21 eV/atom, which is the direction and roughly the size of the known reduction of LDA's overbinding (published values are near 5.3 and 4.6, with experiment at 4.63).

Limitation. The radial atomic solver is LDA-only, so the basis functions and the free-atom reference density of a PBE run are generated from an LDA-PW92 atom. That is ordinary practice — the basis is a variational space, and the crystal is solved self-consistently in the functional actually requested — and the engine states it in the run log rather than leaving it to be discovered.

6 Brillouin-zone integration

6.1 Bloch sums and the generalised eigenproblem

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For a periodic system the basis functions are Bloch-summed over lattice translations,

$$\chi_i^{\mathbf{k}}(\mathbf{r}) \;=\; \sum_{\mathbf{T}} e^{i\mathbf{k}\cdot\mathbf{T}}\,\phi_i(\mathbf{r}-\boldsymbol{\tau}_i-\mathbf{T}). \tag{29}$$

The sum over $\mathbf{T}$ is *finite*, not truncated: ϕ is exactly zero beyond its confinement radius. With $\mathbf{k}$ in fractional coordinates and $\mathbf{T}$ in integer lattice multiples the phase is simply $e^{2\pi i\,(\mathbf{k}\cdot\mathbf{n})}$, which removes the commonest sign-and-factor error in a k-point loop.

The resulting problem is the complex Hermitian generalised eigenproblem

$$H(\mathbf{k})\,C(\mathbf{k}) \;=\; S(\mathbf{k})\,C(\mathbf{k})\,E(\mathbf{k}). \tag{30}$$

It is solved by **canonical orthogonalisation**, not by the Cholesky reduction that a standard routine would use, and the difference matters for this basis. Atomic orbitals on nearby atoms are not linearly independent: squeeze two atoms together, or add diffuse functions, and S acquires eigenvalues at the 10^{-6} level and below. Cholesky of such an S factorises a numerically singular matrix and the resulting orbitals are noise amplified by $1/\sqrt{s}$. Canonical orthogonalisation instead diagonalises $S=UsU^\dagger$, *discards* the directions with $s<\eta\,s_{\max}$, and works in the surviving subspace:

$$X = Us^{-1/2}\big|_{\text{kept}},\qquad H' = X^\dagger H X,\qquad C = XC'. \tag{31}$$

The number of discarded directions is reported, because a nonzero count is a basis problem rather than an eigensolver problem and this is the only place that can say so.

A Hermitian $A+iB$ with A symmetric and B antisymmetric is solved through the real embedding $\left(\begin{smallmatrix}A & -B\\ B & A\end{smallmatrix}\right)$, whose spectrum is that of the Hermitian matrix with each eigenvalue twice. This costs a factor of eight in a step that is not the bottleneck and avoids a separate complex eigensolver to write, test, and get wrong.

6.2 Monkhorst–Pack meshes

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The sampling is a Γ-centred Monkhorst–Pack mesh [8],

$$\mathbf{k}_{pqr} = \frac{p}{N_1}\mathbf{b}_1 + \frac{q}{N_2}\mathbf{b}_2 + \frac{r}{N_3}\mathbf{b}_3, \tag{32}$$

with uniform weights $1/(N_1N_2N_3)$ summing to one. For a finite system the only k-point is Γ, and that is not a choice.

Occupations come from a Fermi–Dirac distribution with the chemical potential found by bisection on the electron count. This is not a convenience for metals: it is required for convergence at all. An open shell — silicon's $3p^2$ is two electrons in a threefold degenerate level — has arbitrary eigenvectors within the degenerate subspace, and filling states one at a time hands both electrons to whichever vector the eigensolver returned first. That choice rotates every iteration: the energy stands still to eight digits while the density swings by more than an electron, indefinitely. A Fermi distribution gives equal occupation to equal eigenvalues by construction.

6.3 Irreducible-zone reduction

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Symmetry-equivalent k-points are found and computed once. The algorithm is exact integer arithmetic throughout, which means orbit membership is decided by comparison and not by a tolerance:

1. The **lattice** point group: every integer matrix W with $W^\top GW = G$, $G_{ij} = \mathbf{a}_i \cdot \mathbf{a}_j$ the metric. That condition says precisely that the corresponding Cartesian map is orthogonal *and* takes the lattice to itself. Candidates are generated from the observation that a symmetry sends each lattice vector to a lattice vector of the same length, so the search is a product of three short lists rather than a sweep over integer matrices. The images are the **columns** of W: with $\mathbf{p}' = W\mathbf{p}$ on fractional coordinates, $\mathbf{a}_j$ has coordinates $\mathbf{e}_j$ and its image is $W\mathbf{e}_j$.

2. The **crystal** point group: those W for which some translation $\mathbf{t}$ makes $\mathbf{p} \to W\mathbf{p} + \mathbf{t}$ map the atoms onto themselves, species by species. Candidate translations are complete and few, since any valid one must carry the first atom onto some atom of its own species.

3. In reciprocal space $\mathbf{k} \to W^{-\top}\mathbf{k}$, which follows from requiring $\mathbf{k} \cdot \mathbf{p}$ invariant, and is again an integer matrix because $\det W = \pm 1$.

4. Time reversal adds $\mathbf{k} \to -\mathbf{k}$, valid for a real Hamiltonian.

5. Operations that do not map the *mesh* onto itself are discarded. A $4 \times 4 \times 2$ mesh does not have the symmetry of its own crystal, and folding with an operation that carries a mesh point off the mesh would silently mis-weight the sum.

Each irreducible point carries $w_\mathbf{k} = n_\mathbf{k}/(N_1N_2N_3)$ with $n_\mathbf{k}$ its orbit size, and $\sum w = 1$ exactly.

Two reductions, and only one of them is exact. Folding the k-set changes the *density*, not only the band energy. Summing $w_\mathbf{k}|\psi_\mathbf{k}|^2$ over a wedge does not reproduce the full-zone sum, because the orbit members contribute $|\psi_\mathbf{k}(\mathcal{R}^{-1}\mathbf{r})|^2$ and not $|\psi_\mathbf{k}(\mathbf{r})|^2$; point-group folding is exact only if the density is symmetrised afterwards, which is not yet implemented.

Time reversal is the exception, and it is exact as it stands: $|\psi_{-\mathbf{k}}|^2 = |\psi_\mathbf{k}^*|^2 = |\psi_\mathbf{k}|^2$ *pointwise* for a real Hamiltonian, so no symmetrisation is needed. It is therefore the default. Measured on silicon, everything else held fixed:

Folding	Points ($4 \times 4 \times 4$)	ΔE vs. full mesh
none	64	—
time reversal	36	-1.1×10^{-5} eV
crystal point group	8	$+8.7 \times 10^{-2}$ eV

The point-group column is left available rather than removed, because it is exact for *eigenvalues* — it is the right setting for a band structure evaluated on an already-converged density — and the engine says so at run time rather than leaving the distinction to this document.

Verification. Group orders and irreducible-wedge sizes are published, checkable numbers, and they are checked: 48 operations for the face-centred cubic lattice and for diamond on it, falling to 8 when one atom is displaced; $4 \times 4 \times 4$ meshes reducing to 8, 8, 10 and 18 points for diamond, face-centred cubic, simple cubic and the displaced case. These agreed with an independent symmetry analyser, and they caught a real defect: building the lattice-vector images as *rows* instead of columns yields the transposed group — the same size, still a group, and completely invisible for a crystal with one atom at the origin, where every operation maps the single position onto itself regardless. Both single-atom test cases passed while diamond was wrong.

7 Self-consistency

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The Kohn–Sham problem is a fixed point: a density gives a potential, the potential gives orbitals, the orbitals give a density. One cycle is

$$\rho \;\rightarrow\; v_{\text{es}}[\rho] + v_{xc}[\rho] \;\rightarrow\; H(\mathbf{k}), S(\mathbf{k}) \;\rightarrow\; \{\varepsilon_{n\mathbf{k}}, C_{n\mathbf{k}}\} \;\rightarrow\; f_{n\mathbf{k}} \;\rightarrow\; \rho^{\text{out}}. \tag{33}$$

The starting density is the superposition of free-atom densities, which is a good guess: most of the density of a solid *is* its atoms, and the difference is what the SCF has to find.

7.1 Mixing

Feeding the output straight back diverges for anything metallic — long wavelength charge sloshes between regions with a gain above one — so each iteration is fed a mixture. Linear mixing is $\rho^{k+1}_{\text{in}} = \rho^k_{\text{in}} + \alpha(\rho^k_{\text{out}} - \rho^k_{\text{in}})$. Pulay/DIIS [9] treats the residual $R^k = \rho^k_{\text{out}} - \rho^k_{\text{in}}$ as an error vector and chooses the combination of the last m inputs whose residuals cancel best,

$$\min_{\{c_i\}} \left\| \sum_i c_i R^i \right\|^2 \quad \text{subject to} \quad \sum_i c_i = 1, \tag{34}$$

solved as the bordered linear system $\begin{pmatrix} B & \mathbf{1} \\ \mathbf{1}^\top & 0 \end{pmatrix}\begin{pmatrix} c \\ \lambda \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ with $B_{ij} = R^i \cdot R^j$. The B matrix becomes singular *exactly* at convergence, when the residuals are all nearly zero and therefore nearly parallel; the implementation detects that and falls back to a linear step rather than throwing away a converged answer on the last iteration.

7.2 Convergence criteria and the total energy

Both the energy change and the integrated density residual $\int |\rho^{\text{out}} - \rho^{\text{in}}| d\mathbf{r}$ must fall below their thresholds. Requiring both is deliberate: the energy is variational and therefore second order in the density error, so it settles while the density is still moving. A loop that stops on energy alone stops early and reports a density nobody converged.

The total energy is assembled as

$$E = \underbrace{\sum_{n\mathbf{k}} f_{n\mathbf{k}}\varepsilon_{n\mathbf{k}} - \int \rho\, v_{\text{eff}}}_{T_s} + E_{\text{es}} + E_{xc}. \tag{35}$$

During the loop the electrostatic and exchange–correlation terms are evaluated on the *input* density (the Harris–Foulkes form), because building v_{es} is the most expensive step in an iteration; the exact Kohn–Sham functional is evaluated once on the converged density at the end.

8 Energy derivatives

8.1 Forces

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Because NAOs are *atom-centred and move with the nuclei*, the force has two parts and omitting the second produces forces inconsistent with the energy surface. The **Hellmann–Feynman** term is the electrostatic force on the bare nucleus,

$$\mathbf{F}_A^{\text{HF}} = Z_A \int \rho(\mathbf{r}) \frac{\mathbf{r} - \mathbf{R}_A}{|\mathbf{r} - \mathbf{R}_A|^3} \mathrm{d}\mathbf{r} - Z_A \sum_{B \neq A} Z_B \frac{\mathbf{R}_A - \mathbf{R}_B}{|\mathbf{R}_A - \mathbf{R}_B|^3}, \tag{36}$$

and the **Pulay** term arises because $\partial\phi_i/\partial\mathbf{R}_A \neq 0$:

$$\mathbf{F}_A^{\text{Pulay}} = -\sum_{ij} \left[D_{ij} \frac{\partial H_{ij}}{\partial \mathbf{R}_A} - W_{ij} \frac{\partial S_{ij}}{\partial \mathbf{R}_A} \right], \tag{37}$$

with $W_{ij} = \sum f\varepsilon\, C_i^* C_j$ the energy-weighted density matrix.

What is implemented, and what actually drives a relaxation. The production force is the **finite difference** $-\mathrm{d}E/\mathrm{d}\mathbf{R}_A$, which is exact for any functional and either boundary condition because it differences the same total energy the SCF minimises. It costs $6N$ self-consistency cycles and it is what the optimiser uses. Of the analytic decomposition, the Hellmann–Feynman term and the *overlap* part of Pulay are implemented; $\sum_{ij} D_{ij}\partial H_{ij}/\partial\mathbf{R}_A$ and the derivatives of the Becke weights are not, and on a stretched H_2 bond the implemented terms give 5.2 eV/Å where the true force is 0.79 — the missing terms dominate. Both are reported so the gap is visible rather than assumed away.

A finite-difference force is only as smooth as the energy. This nearly sank the whole section, and the diagnosis is worth recording. The force initially carried between 1 and 100 eV/Å of noise, with no plateau in the displacement and no improvement on grid refinement. It was not the force: the *energy surface* was not converged. The neutral-atom pair energy $\int \rho_A v_B^{\text{NA}}$ was being evaluated on atom A's spherical grid, and v_B^{NA} carries the nuclear pole of B — a singularity sitting off-centre, exactly what the multicentre partition exists to prevent. It is invisible for a single atom, which has no pairs at all and reproduced the radial solver to 0.001 eV, and fatal for two.

The cure is to do that integral exactly. Both factors are spherical about their own centres, so the angular average of v_B over a sphere of radius r about A, the centres d apart, has a closed form and the pole cancels analytically:

$$\langle v_B \rangle(r) = \frac{1}{2rd}\int_{|r-d|}^{r+d} v_B(t)\, t\mathrm{d}t, \qquad \int \rho_A v_B = \frac{2\pi}{d}\int \rho_A(r)\, r\big[C(r+d) - C(|r-d|)\big]\mathrm{d}r, \tag{38}$$

with $C(t) = \int_0^t v_B t'\mathrm{d}t'$ and $v_B(t)\, t = \mathrm{screened}(t)\, t - Z_B$ smooth through the origin. This is a one-dimensional integral on the radial mesh with no three-dimensional grid involved. The angular drift of $\mathrm{E}(\mathrm{H}_2)$ fell from 0.105 eV to 0.0007 eV, and the force went from noise to a clean plateau.

Guard. ‘finiteDifference‘ evaluates at h and $2h$ and returns a numerical failure when they disagree. An exact derivative barely moves; grid noise halves. That guard is what caught the pole, and it stays.

8.2 Stress tensor

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From the response of the total energy to a uniform strain, $\mathbf{x} \to (\mathbb{1} + \boldsymbol{\varepsilon})\mathbf{x}$ applied to the lattice vectors *and* the atomic positions:

$$\sigma_{\alpha\beta} = \frac{1}{\Omega}\frac{\partial E}{\partial \varepsilon_{\alpha\beta}}. \tag{39}$$

Six independent components by central differences. Every term of the energy contributes automatically — the matrices through the strained inter-atomic vectors, the electrostatics through the lattice sum and the multipole expansion, the exchange–correlation through the volume element, and the basis through the same mechanism that produces the Pulay force — because all of them sit inside the energy being differenced. An off-diagonal component perturbs $\varepsilon_{\alpha\beta}$ and $\varepsilon_{\beta\alpha}$ together, so its derivative carries a factor of two that is divided back out.

Verification. Cubic symmetry is the sharpest available check and needs no reference number, because nothing in the calculation is told the crystal is cubic: for silicon the off-diagonal components come out at 5×10^{-10} eV/Å^3 and the three diagonal components agree to 1.6×10^{-9}. The magnitude is checked independently against a volume scan — the trace gives 55.9 GPa against 52.5 GPa from $-\mathrm{d}E/\mathrm{d}V$, agreeing inside the stress's own noise estimate.

A strained cell is harder to converge than the one it came from. Coarse stress settings failed on self-consistency iterations rather than on grid quality; gentler mixing and a higher ceiling fixed it outright. Worth knowing before blaming the strain derivative.

8.3 Geometry optimisation

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Relaxation uses FIRE — damped dynamics that removes the damping while the system descends and freezes the velocity the moment it starts to climb. It is chosen over a quasi-Newton method because it needs no line search: a single force evaluation costs $6N$ self-consistency cycles, and a rejected line-search trial would throw all of them away.

H_2 started at 0.95 Å relaxes monotonically to 0.87 Å with the largest force falling from 1.76 to 0.043 eV/Å in twenty steps. The residual difference from the LDA reference bond length of 0.77 Å is the confined minimal basis, not the force.

9 Validation

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An all-electron total energy that has not been checked against something is a number with no standing. Each layer of the engine is therefore checked against something that is *not* another layer.

Quantity	Checked against	Agreement
Radial solver, $v_{\mathrm{el}} = 0$	analytic $-Z^2/2n^2$	10^{-10} rel.
Angular rules	$\int Y_{\ell m} Y_{\ell' m'} \mathrm{d}\Omega = \delta$	machine prec.
Eigensolver	constructed $QDQ^{\top}$ spectrum; LAPACK	10^{-12}
Si atom, E_{tot}	grid-converged, 4× mesh refinement	10^{-8} Ha
Si atom eigenvalues	independent all-electron atomic solver	see below
3D pipeline	the engine's own 1D radial solver	0.08 eV
Cell electron count	exact integer	$< 10^{-7}$
Translational invariance	same energy for a shifted cell	0.003 meV

The silicon atom. LDA-PW92 gives $E_{\mathrm{tot}} = -288.19374$ Ha with $\varepsilon_{1s} = -65.18430$ and $\varepsilon_{3p} = -0.15331$ Ha, grid-converged to 10^{-8} across a fourfold mesh refinement. An independent all-electron atomic solver converges *onto* these values as its own radial grid is refined (ε_{1s}: -65.18170 at its default mesh, -65.18429 at 7500 points).

The three-dimensional pipeline against the one-dimensional one. Running the free silicon atom through the full multicentre machinery reproduces the radial answer to 0.08 eV, and does so *from above* — the confined basis is a restriction of the variational space, so its minimum cannot lie below the exact one. A 3D energy underneath the 1D answer would mean the quadrature is manufacturing binding.

Measured effect of the confinement barrier. Section 2.4 argued that a hard wall leaves an unrepresented surface term in every inter-atomic kinetic matrix element. The cohesive energy of silicon, everything else held fixed, shows it:

Basis	Hard wall	Smooth barrier
Tier 1 (minimal)	7.11	**5.68**
Tier 2 (DZ + polarisation)	10.77	**6.26**
Tier 3	66.56	**6.41**

eV/atom, $2 \times 2 \times 2$ mesh. Converged LDA is ≈ 5.3 eV/atom; experiment is 4.63.

The wall column diverges as the basis grows — each added function brings its own cutoff sphere and its own missing surface term. The barrier column is a proper convergence sequence, with increments of +0.57 then +0.15 eV. What remains above the reference is accounted for by the spherical, non-spin-polarised atomic reference (worth roughly 0.6 eV for silicon), residual confinement, and a k-mesh that is not yet converged (Section 6.3).

What is not validated. The absolute total energy of a *periodic* system has no external reference available: a pseudopotential code's total energy is not comparable with an all-electron one. The periodic path is therefore checked on what can be checked — electron count, self-consistency, degeneracies required by symmetry, translational invariance — and the absolute energy is reported without being asserted.

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